Lecture 07 — Norm, distance, RMS, and variation

Fri Sep 11

Last updated: September 11, 2026.

Boyd video (optional): Lecture 9 — norm · Lecture 10 — distance · Lecture 11 — standard deviation · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 3.1–3.3 (norm, distance, standard deviation). Quiz 3 prep: Boyd 3.1, 3.2, 3.15 and distance computations. Related objectives: Lectures. Schedule. Quiz 3 is Friday, September 18. Angle and cosine similarity are next class (Lecture 08).

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples (hand and software), and **Check yourself**.

Learning objectives

By the end of class, students should be able to:

1. Compute vector norms and distances.
2. Interpret norm and distance geometrically and contextually.
3. Relate norm to RMS and, where appropriate, variation around an average.
4. Use software to calculate distances in dimensions too large for hand computation.

Warm-up

Let $u = (3, -4)$ and $v = (0, 5)$.

(a) Compute $u^\mathsf{T}u$ and $\sqrt{u^\mathsf{T}u}$ by hand.

(b) Without a calculator, which is larger: $\|u\|$ or $\|v\|$? Brief reason.

Fill in

The **Euclidean norm** of an n-vector x is

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2} = \sqrt{_____}.$$

It measures the _____ of x. For a scalar, $\|x\|$ is the _____.

Key properties (same size vectors; β a scalar): - $\|\beta x\| =$ _____ $\|x\|$ - $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality) - $\|x\| \geq 0$, and $\|x\| = 0$ only when $x =$ _____

The **RMS** (root-mean-square) value is

$$\mathrm{rms}(x) = \frac{\|x\|}{\sqrt{n}} = \sqrt{\frac{x_1^2 + \cdots + x_n^2}{n}}.$$

It describes a _____ size of an entry, and is useful when comparing vectors of different _____.

The **distance** between n-vectors a and b is

$$\mathrm{dist}(a, b) = \|a - b\|.$$

Geometrically this is the length of the _____ from a to b. Two vectors are "close" when this distance is _____ (relative to the application).

The **average** is $\mathrm{avg}(x) = \frac{1}{n}\mathbf{1}^{\mathsf{T}}x$. The **de-meaned** vector is $\tilde{x} = x - \mathrm{avg}(x)\mathbf{1}$. The **standard deviation** is

$$\mathrm{std}(x) = \mathrm{rms}(\tilde{x}) = \frac{\|x - \mathrm{avg}(x)\mathbf{1}\|}{\sqrt{n}},$$

which measures typical _____ of the entries from their average.

Boyd's identity: $\mathrm{rms}(x)^2 = \mathrm{avg}(x)^2 + \mathrm{std}(x)^2$.

Software (keep this page open)

In this course we use either Octave or Python for norms and distances.

Octave (column vectors are fine):

```
x = [3; -4; 0];
norm(x)                 % Euclidean norm
sqrt(x'*x)              % same idea via the inner product
y = [1; 2; -1];
norm(x-y)               % distance ||x-y||
```

Python (NumPy):

```
import numpy as np
x = np.array([3.0, -4.0, 0.0])
np.linalg.norm(x)       # Euclidean norm
np.sqrt(x @ x)          # same idea via the inner product
y = np.array([1.0, 2.0, -1.0])
np.linalg.norm(x - y)   # distance ||x-y||
```

Handy formulas (either language): - $\mathrm{rms}(x) = \|x\|/\sqrt{n}$ - $\mathrm{avg}(x) = \frac{1}{n}\sum_i x_i$ (Octave `mean(x)`, Python `np.mean(x)`) - $\mathrm{std}(x)$ in Boyd's sense is the RMS of the de-meaned vector (population form with divisor n, not $n-1$)

Worked in class

W1. Norm by hand and by software (Lachniet-style). Let

$$u = (2,\ -4,\ 0).$$

(a) Compute $\|u\|$ by hand using $\sqrt{u^{\mathsf{T}}u}$.

(b) Write the Octave command and the Python command that compute the same value. What should both print?

(c) Form the **unit vector** $\hat{u} = u/\|u\|$. What is $\|\hat{u}\|$?

W2. Distance and nearest neighbor (Quiz 3 / Boyd 3.10–3.11 style). Arcata morning temperatures (°F) for three days:

$$a = (52, 51, 53), \qquad b = (54, 50, 55), \qquad c = (60, 58, 59).$$

A new day is $x = (53, 52, 54)$.

(a) Compute $\|x - a\|$, $\|x - b\|$, and $\|x - c\|$ by hand or with software.

(b) Which day is the nearest neighbor of x?

(c) In one sentence, what does "nearest" mean here?

W3. Boolean distance (Boyd 3.1). Suppose x and y are Boolean n-vectors (entries only 0 or 1).

(a) Show that $\|x - y\|^2$ equals the number of positions where x and y disagree.

(b) Therefore $\|x - y\| = \sqrt{k}$ where k is that disagreement count. For $x = (1, 0, 1, 1, 0)$ and $y = (1, 1, 0, 1, 0)$, find k and $\|x - y\|$.

W4. RMS, average, and standard deviation (Boyd 3.15 setup). Let

$$x = (2, 4, 4, 6).$$

(a) Compute $\text{avg}(x)$, $\|x\|$, $\text{rms}(x)$, the de-meaned vector $\tilde{x}$, and $\text{std}(x)$.

(b) Check Boyd's identity $\text{rms}(x)^2 = \text{avg}(x)^2 + \text{std}(x)^2$.

(c) Interpret $\text{std}(x)$ in one short sentence for this data.

W5. Feature scale changes nearest neighbor (Quiz 3 C1.2 style). Two features: area (in 1000 ft^2) and bedrooms. Houses:

$$h_1 = (1.6,\ 2), \qquad h_2 = (1.5,\ 2), \qquad h_3 = (1.6,\ 4).$$

(a) Compute $\|h_1 - h_2\|$ and $\|h_1 - h_3\|$. Which house is nearer to h_1?

(b) Now measure area in square feet instead, so the vectors become

$$\tilde{h}_1 = (1600,\ 2), \qquad \tilde{h}_2 = (1500,\ 2), \qquad \tilde{h}_3 = (1600,\ 4).$$

Compute $\|\tilde{h}_1 - \tilde{h}_2\|$ and $\|\tilde{h}_1 - \tilde{h}_3\|$. Which house is nearer to $\tilde{h}_1$ now?

(c) What changed, and why does Boyd warn about units on heterogeneous features?

Check yourself

C1. Let $w = (1, -2, 2)$. Compute $\|w\|$ by hand. Then write one Octave line and one Python line that compute it.

C2. Quiz 3 C1.1 style. Observation $x = (1, 2)$. Candidates:

$$z_1 = (0, 2), \quad z_2 = (3, 2), \quad z_3 = (1, 5), \quad z_4 = (2, 1), \quad z_5 = (1, 2.5).$$

Compute the five distances $\|x - z_i\|$ (hand or software). Which candidate is nearest?

C3. Boyd 3.2 style. Let a be an n-vector and b an m-vector, and set $x = (a, b)$ (stack). Express $\text{rms}(x)$ in terms of $\text{rms}(a)$, $\text{rms}(b)$, n, and m.

C4. For $t = (10, 10, 10, 14)$, find $\text{avg}(t)$ and $\text{std}(t)$. Use Boyd's definition (RMS of the de-meaned vector).

C5. True or false, with one sentence: if $\|x\| = \|y\|$, then $x = y$ or $x = -y$.

Answers (Check yourself)

C1. $\|w\| = \sqrt{1+4+4} = 3$. Octave: `norm([1;-2;2])`. Python: `np.linalg.norm(np.array([1.,-2.,2.]))`.

C2. Distances: 1, 2, 3, $\sqrt{2} \approx 1.414$, 0.5. Nearest is $z_5 = (1, 2.5)$.

C3. $\text{rms}(x) = \sqrt{\dfrac{n\,\text{rms}(a)^2 + m\,\text{rms}(b)^2}{n+m}}$.

C4. $\text{avg}(t) = 11$. De-meaned $(-1, -1, -1, 3)$; $\text{std}(t) = \sqrt{(1+1+1+9)/4} = \sqrt{3}$.

C5. **False.** Example: $x = (1, 0)$ and $y = (0, 1)$ have the same norm but are neither equal nor negatives.